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Securing a computing device accessory

Abstract

An online service may use a computing system to authenticate a remote accessory device by establishing an encrypted communication channel with the remote accessory device, the remote accessory device being connected to a host device. Then, using the encrypted communication channel, obtain a device certificate possessed by the remote accessory device. The online service may then receive a first device token from the remote accessory device and determine whether the first device token exists within a token history associated with the device certificate. Based on the first device token existing within the token history, the online service may then determine that the remote accessory device is an authentic device; generate a second device token for the remote accessory device; store the second device token in the token history associated with the device certificate; and send the second device token to the remote accessory device.

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Background/Summary

BACKGROUND

(1) An increase of counterfeit computing device accessories has caused problems for legitimate accessory manufacturers and those who sell legitimate devices. To prevent this issue, numerous verification and authorization techniques have been developed to authenticate legitimate accessories and ban or block counterfeit accessories.

(2) An example of an authentication technique is to include a security chip in each authentic device. This security chip may be utilized to locally authenticate the accessory with a host device. While this provides a good starting point, if the chips' information (e.g., cryptographic key, security certificate) is compromised, counterfeit devices may get past this authentication step.

(3) As such, there is an ongoing need to provide verification methods to authorize legitimate computing device accessories and identify counterfeit computing device accessories. While the use of security chips provides a first authentication step, a security chip may be compromised allowing counterfeit devices to mimic authentic devices.

(4) The subject matter claimed herein is not limited to embodiments that solve any disadvantages or that operate only in environments such as those described above. Rather, this background is only provided to illustrate one example technology area where some embodiments described herein may be practiced.

BRIEF SUMMARY

(5) In some aspects, the techniques described herein relate to a method, implemented at a computer system that includes a processor, including: establishing an encrypted communication channel with a remote accessory device that is connected to a host device; receiving a first device token from the remote accessory device; determining that the first device token exists within a token history associated with a device certificate of the remote accessory device; determining that the remote accessory device is an authentic device, based on the first device token existing within the token history; generating a second device token for the remote accessory device; storing the second device token in the token history associated with the device certificate of the remote accessory device; and sending the second device token to the remote accessory device.

(6) In some aspects, the techniques described herein relate to a method, implemented at a computer system that includes a processor, for banning a remote accessory device, the method including: establishing an encrypted communication channel with a remote accessory device, the remote accessory device being connected to a host device; using the encrypted communication channel, obtaining a device certificate possessed by the remote accessory device; receiving a first device token from the remote accessory device; determining whether the first device token exists within a token history associated with the device certificate; and based on the first device token not existing within the token history, determining that the remote accessory device is counterfeit; and adding the device certificate to a ban list.

(7) In some aspects, the techniques described herein relate to a computer system for authenticating a remote accessory device, including: a processing system; and a computer storage media that stores computer-executable instructions that are executable by the processing system to at least: obtain, over an encrypted network, a device certificate possessed by a remote accessory device; receive a first device token from the remote accessory device; determine whether the first device token exists within a token history associated with the device certificate; and based on the first device token existing within the token history, determine that the remote accessory device is an authentic device; generate a second device token for the remote accessory device; store the second device token in the token history associated with the device certificate; and send the second device token to the remote accessory device.

(8) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe the manner in which the advantages and features of the systems and methods described herein can be obtained, a more particular description of the embodiments briefly described above will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments of the systems and methods described herein, and are not therefore to be considered to be limiting of their scope, certain systems and methods will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. 1A illustrates an example computer architecture that facilitates authorization of a remote accessory device;

(3) FIG. 1B illustrates an example of internal elements of an authentication component;

(4) FIG. 2 illustrates an example of internal components of a host device;

(5) FIG. 3 illustrates an example of internal components of a remote accessory device;

(6) FIG. 4 illustrates an example diagram of authorizing a remote accessory device; and

(7) FIG. 5 illustrates a flow chart of an example method for actions based on authenticated or counterfeit remote accessory devices.

DETAILED DESCRIPTION

(8) As the number of counterfeit remote accessory devices have increased, authenticating remote accessory devices from legitimate accessory manufacturers has turned into an ongoing challenge. In particular, counterfeit device manufacturers may utilize traditional authentication techniques (e.g., using stolen certificates), creating an ongoing need for new and improved ways to authenticate legitimate devices.

(9) Previously, these challenges have been dealt with by utilizing a security chip to locally authenticate the remote accessory device with a host device. However, in some instances, the security chip may become comprised. In these instances, the counterfeit device may use a counterfeit security chip and the counterfeit device is authenticated by using a security chip authentication technique. Additionally, there may be security chip shortages resulting in a lack of security chips in legitimate devices. Therefore, there is a need for authentication techniques that do not rely solely on security chips.

(10) At least some embodiments described herein remotely authenticate an accessory device. These embodiments therefore add an additional layer of authentication (e.g., a remote server-based authentication) to ensure authentic devices are registered and counterfeit devices are added to a ban list. Disclosed embodiments add an additional layer of security to devices made by device manufacturers to determine the device's legitimacy without relying on security chips which are easily compromised or unavailable.

(11) FIG. 1A illustrates an example computer architecture **100a** that facilitates authentication of a remote accessory device. As shown, computer architecture **100a** includes an online service **101** comprising processing system(s) **102** (e.g., a single processor, or a plurality of processors), memory **103** (e.g., system or main memory), storage media **106** (e.g., a single computer-readable storage medium, or a plurality of computer-readable storage media), all interconnected by a bus **105**. As shown, in embodiments, online service **101** also includes a network interface **104** (e.g., one or more network interface cards) for interconnecting (via a network **111**; e.g., a wide area network, such as the Internet) to a host device **112** and/or a remote accessory device **113**. The host device **112** and the remote accessory device **113** are also connected either locally or remotely. As examples, in embodiments a local connection is a physical wired connection, a personal area network (PAN) connection such as a BLUETOOTH wireless connection, or a local area network (LAN) connection such as a WIFI wireless connection. As an example, in embodiments a remote connection is a wide area network (WAN) connection, such as using network **111**.

(12) The storage media **106** is illustrated as storing computer-executable instructions implementing at least an authentication component **110** for authenticating the remote accessory device **113**. The storage media **106** is illustrated as also storing data such as a certificate database **107**, a token history **108**, and a ban list **109**.

(13) In embodiments, computer architecture **100a** enables the remote accessory device **113** to be authenticated by the online service **101** via the network **111**. In some embodiments, the remote accessory device **113** is pre-authenticated by the host device **112**. In embodiments, the pre-authentication occurs automatically when the remote accessory device **113** is connected to the host device **112**, or manually by a user request.

(14) FIG. 1B illustrates an example **100b** of internal elements of the authentication component **110** of FIG. 1A. Each internal component of the authentication component **110** depicted in FIG. 1B represents various functionalities that the authentication component **110** might implement in accordance with various embodiments described herein. It will be appreciated, however, that the depicted components—including their identity and arrangement—are presented merely as an aid in describing example embodiments of the authentication component **110**.

(15) The authentication component **110** depicted in FIG. 1B includes a communication component **114**, a device certificate component **115**, a device token component **116**, and a token generator **117**. In embodiments, the communication component **114** enables the online service **101** to establish an encrypted connection with the host device **112** and/or the remote accessory device **113**. In embodiments, the device certificate component **115** enables the online service **101** to request and receive a device certificate from the remote accessory device **113**, and to verify the validity of certificates. In embodiments, the device token component **116** enables online service **101** to request and receive a device token from the remote accessory device **113**, and to verify the validity of device tokens. In embodiments, the token

generator **117** enables the online service **101** to generate new device tokens and send the device token to the remote accessory device **113**.

(16) FIG. **2** illustrates an example **200** of internal components of a host device **201**, such as the host device **112** of FIG. **1A**. Each internal component of the host device **201** depicted in FIG. **2** represents various functionalities that the host device **201** might implement in accordance with various embodiments described herein. It will be appreciated, however, that the depicted components—including their identity and arrangement—are presented merely as an aid in describing example embodiments of the host device **201**.

(17) The host device **201** depicted in FIG. **2** includes processing system(s) **202** (e.g., a single processor, or a plurality of processors), memory **203** (e.g., system or main memory), storage media **206** (e.g., a single computer-readable storage medium, or a plurality of computer-readable storage media), all interconnected by a bus **205**. As shown, the host device **201** may also include a network interface **204** (e.g., one or more network interface cards) for interconnecting (via network **111**) to the online service **101**. The host device **201** may also connect locally (e.g., physical wired connection, PAN, LAN) or remotely (e.g., WAN) to the remote accessory device **113**.

(18) The storage media **206** is illustrated as storing computer-executable instructions implementing at least a pre-authentication component **207** for pre-authenticating the remote accessory device **113**. In embodiments, the pre-authentication component **207** authenticates the remote accessory device **113** based on a security chip at the remote accessory device **113**. In some embodiments, the pre-authentication component **207** determines if a device certificate (e.g., stored within a security chip) associated with the remote accessory device **113** is valid. To determine if the device certificate is valid, in some embodiments, the pre-authentication component **207** compares the device certificate to a known device certificate list. In some embodiments, the pre-authentication is performed locally and the device certificate list is stored in the storage media **206** of the host device **201**. In other embodiments, the pre-authentication is performed remotely and the host device **201** accesses a certificate list from the storage media **106** of the online service **101**. In embodiments, the remote accessory device **113** lacks a device certificate or a security chip, and the remote accessory device **113** is unable to be used with the host device **112** immediately, therefore identifying counterfeit devices quickly and efficiently.

(19) FIG. **3** illustrates an example **300** of internal components of a remote accessory device **301**, such as the remote accessory device **113** of FIG. **1A**. Each internal component of the remote accessory device **301** depicted in FIG. **3** represents various functionalities that the remote accessory device **301** might implement in accordance with various embodiments described herein. It will be appreciated, however, that the depicted components—including their identity and arrangement—are presented merely as an aid in describing example embodiments of the remote accessory device **301**.

(20) The remote accessory device **301** depicted in FIG. **3** includes processing system(s) **302** (e.g., a single processor, or a plurality of processors), memory **303** (e.g., system or main memory), storage media **306** (e.g., a single computer-readable storage medium, or a plurality of computer-readable storage media), all interconnected by a bus **305**. As shown, the remote accessory device **301** may also include a network interface **304** (e.g., one or more network interface cards) for interconnecting (via network **111**) to the online service **101**. The remote accessory device **301** may also connect locally or remotely to the host device **112**.

(21) The storage media **306** is illustrated as storing data associated with the remote accessory device **301**. For example, FIG. **3** illustrates the storage media **306** as storing a device certificate **307**, a first device token **308**, and a second device token **309**.

(22) FIG. **4** illustrates an example authentication diagram showing a method **400** for authenticating a remote accessory device **113**. The authentication diagram includes an online service **400a** (e.g., online service **101**), a host device **400b** (e.g., host device **112**), and a remote accessory device **400c** (e.g., remote accessory device **113**). As an example, the online service **400a** is an online gaming service (e.g., Xbox Live, PlayStation Network, Nintendo Switch Online), the host device **400b** is a gaming console (e.g., an Xbox, a PlayStation, a Nintendo Switch), and the remote accessory device **400c** is a gaming controller (e.g., an Xbox controller, a PlayStation controller, a Nintendo Switch controller). In some embodiments, the gaming console is a virtual gaming console. In embodiments, instructions for implementing method **400** are encoded as computer-executable instructions (e.g., authentication component **110**, pre-authentication component **207**) stored on a computer storage media that are executable by a processing system to cause online service **400a**, host device **400b**, and/or remote accessory device **400c** to perform the method.

(23) Referring now to specifics of method **400**, initially the remote accessory device **400c** and host device **400b** connect to one another. In some embodiments the host device is locally connected to the remote accessory device. In other embodiments, the host device is remotely connected to the remote accessory device. For example, the remote accessory device **400c** connects to the host device **400b** (act **401**), and the host device **400b** connects to the remote accessory device **400c** (act **402**). The connection may occur wirelessly by using BLUETOOTH, over a wireless network, over a wired connection, or other appropriate connection methods.

(24) Once the remote accessory device **400c** and the host device **400b** are connected (e.g., act **401** and act **402**), the host device **400b** performs a pre-authorization of the remote accessory device **400c** (act **403**). In embodiments, act **403** includes the remote accessory device having been pre-authorized by the host device. The pre-authorization may occur automatically when the host device **400b** and remote accessory device **400c** are connected (e.g., based on act **401** and

act **402**), when requested by a user, or when requested by a third party. In embodiments, the pre-authorization occurs regardless of if the host device **400b** is connected to a network or not. In embodiments, the host device pre-authorizes the remote accessory device based only on communications between the remote accessory device and the host device. In some embodiments, the pre-authentication component **207** determines if the device certificate associated with the remote accessory device **113** is valid. To determine if the device certificate is valid, in some embodiments, the pre-authentication component **207** compares the device certificate to a known device certificate list. In some embodiments, the pre-authentication is performed locally and the device certificate list is stored in the storage media **206** of the host device **201**. In the instance where other authentication methods are temporarily unavailable, the pre-authentication provides an efficient first authentication until other authentication techniques are performed.

(25) Next, the online service **400a** and remote accessory device **400c** establish an encrypted communication channel (act **404** and act **405**). In some embodiments, the encrypted communication channel is established only when the pre-authorization succeeds. In other embodiments, the encrypted communication channel is established whether the pre-authorization succeeds or fails. In some embodiments, the encrypted communication channel is a TLS 1.3 communication channel, though a variety of encryption protocols could be used. In some embodiments (e.g., when the remote accessory device **400c** is capable of communicating over network **111**), the encrypted communication channel is established between the remote accessory device **400c** and the online service **400a** directly. In other embodiments (e.g., when the remote accessory device **400c** is incapable of communicating over network **111**), the encrypted communication channel is facilitated by an intermediary, such as the host device **400b**.

(26) From the perspective of the online service **400a**, in embodiments, act **405** comprises establishing an encrypted communication channel with a remote accessory device, the remote accessory device being connected to a host device. In an example, the communication component **114** establishes an encrypted communications channel with the remote accessory device **400c**, either directly or via the host device **400b**.

(27) In some embodiments, using the encrypted communication channel, the online service **400a** requests a device certificate from the remote accessory device **400c** (act **406**). From the perspective of the online service **400a**, in embodiments act **406** comprises sending, to the remote accessory device, a request for the device certificate. The remote accessory device **400c**, in turn, receives the device certificate request (act **407**) and sends the device certificate to the online service **400a** over the encrypted communication channel (act **408**). In other embodiments, the remote accessory device **400c** sends the device certificate (act **408**) without a request from the online service **400a**.

(28) Whether expressly requested or not, in embodiments the online service **400a** receives the device certificate from the remote accessory device **400c** (act **409**). For example, in some embodiments, the online service **400a** explicitly requests the device certificate from the remote accessory device **400c** (act **406**). In other embodiments, the remote accessory device **400c** sends (act **408**) the device certificate to the online service **400a** without an explicit request. In embodiments, act **409** comprises, using the encrypted communication channel, obtaining a device certificate possessed by the remote accessory device. In an example, the device certificate component **115** receives a device certificate from remote accessory device **400c**.

(29) In embodiments, upon receipt of the device certificate, the online service **400a** verifies that the device certificate is valid. This validation process may include checking the device certificate against a certificate database **107** to determine whether the device certificate **307** exists within the certificate database **107**. The verification process may further include, in some embodiments, checking if the device certificate has previously been used by any remote accessory device. In the case where the device certificate has not been used, the online service **400a** may register the remote accessory device **400c** in an online database. In some embodiments, the database stores a list of registered devices. As an example, a counterfeit device may use a compromised device certificate which has previously been registered. In this case, when the counterfeit device attempts to be registered, the device is quickly identified as counterfeit and added to a ban list.

(30) From the perspective of the online service **400a**, the validation process includes determining the device certificate is valid. In an example, the device certificate is determined valid by the device certificate component **115**. In some embodiments, the validation process comprises determining that the remote accessory device is counterfeit is based on identifying the device certificate on a ban list. In other embodiments, the validation process comprises determining that the remote accessory device is counterfeit is based on the device certificate being neither in the online database nor the ban list.

(31) In some embodiments, once the device certificate has been received by the online service **400a** (act **409**), and the online service **400a** has verified validity of the certificate (act **410**), the online service **400a** sends a request for an authentication token to the remote accessory device **400c** (act **411**). The remote accessory device **400c** receives the request for the authentication token (act **412**), and sends a device token (first token) stored at the remote accessory device **400c** (act **413**). From the perspective of the online service **400a**, act **411** comprises sending, to the remote accessory device, a request for an authentication token and act **414** comprises receiving the first device token from the remote accessory device based on the request. In examples, the device token component **116** requests and receives the device token from the remote accessory device **400c**. In other embodiments, the remote accessory device **400c** sends the device token (first token) to the online service **400a** without receiving an express request from the online service **400a**. Thus, in some embodiments, method **400** lacks act **411** and act **412**. In either case, the online service **400a** receives the

token (act **414**) and determines whether the token exists in a token history (act **415**) maintained by the online service **400a** (e.g., token history **108**). In some embodiments, the remote accessory device **400c** sends the device certificate and the device token to the online service **400a** in a single operation. In some embodiments, the remote accessory device **400c** sends the device certificate and/or device token to the host device **400b**, which acts as an intermediary to forward the device certificate and/or the device token to the online service **400a**.

(32) Continuing from FIG. 4, FIG. 5 illustrates a flow chart of an example method **500**, comprising acts that occur once it is determined if the first device token exists in the token history (act **414**, act **501**). In embodiments, instructions for implementing method **500** are encoded as computer-executable instructions (e.g., authentication component **110**) stored on a computer storage media (e.g., storage media **106**) that are executable by a processing system (e.g., processing system(s) **102**) to cause a computer system (e.g., online service **101**) to perform the method. The following discussion now refers to a number of methods and method acts. Although the method acts may be discussed in certain orders, or may be illustrated in a flow chart as occurring in a particular order, no particular ordering is required unless specifically stated, or required because an act is dependent on another act being completed prior to the act being performed.

(33) In some embodiments, method **500** operates to authenticate a remote accessory device. Referring to act **501**, when the first device token exists in the token history (e.g., the “Yes” branch from act **501**), in embodiments, the remote accessory device is determined to be authentic (act **502**). The online service **101** then generates a second device token (act **503**) and stores the second device token (act **504**) in the device token history (e.g., token history **108**). The online service **101** sends the second device token to the remote accessory device (act **505**). In some embodiments, the online service **101** sends an indication to the host device (act **506**) that the remote accessory device **113** has been authenticated. In embodiments, the indication allows the remote accessory device **113** to continue to be used with the host device **112**.

(34) From the perspective of the online service **400a**, in embodiments based on the first device token existing within the token history (act **501**), act **502** comprises determining that the remote accessory device is an authentic device. As an example, the device token component **116** determines whether the first device token exists in the token history **108**. In embodiments, act **503** comprises generating a second device token for the remote accessory device and storing the second device token in the token history associated with the device certificate. In an example, the token generator **117** generates the second token and also stores the second token in the token history. Act **504**, in embodiments, comprises sending the second device token to the remote accessory device. As an example, the communication component **114** sends the second device token from the online service **400a** to the remote accessory device **400c** either directly or through the host device **400b**. Act **506** comprises sending an indication, to the host device, that the remote accessory device is authentic.

(35) The token history **108** may store the first device token, the second device token, and other generated tokens (e.g., up to n device tokens, where n is a positive integer). In some embodiments, online service **101** deletes the first device token from the token history once the second device token is generated. In other embodiments, the online service **101** deletes tokens from the token history **108** once a set number of token are stored. Other embodiments, the online service **101** periodically delete tokens from the token history **108**. In even other embodiments, the token history **108** may be stored permanently.

(36) In some embodiments, method **500** operates to ban a remote accessory device. Referring again to act **501**, when the first device token does not exist in the token history (e.g., the “No” branch from act **501**), in embodiments, the remote accessory device is determined to be a counterfeit (act **507**). The device certificate is added to a ban list (act **508**) stored by the online service **101** (e.g., ban list **109**). The communication component **114** then sends an indication to the host device **112** (act **509**). In some embodiments, based on this indication, the host device **112** terminates the connection with the remote accessory device (act **510**). In some embodiments, the host device **112** waits to terminate the connection with the remote accessory device **113** until multiple attempts to authenticate the device have been performed. In some embodiments, the online service **101** shares the ban list **109** with multiple devices (e.g., multiple host devices and remote accessory devices). In some embodiments, the online service **101** checks the ban list **109** for the received device certificate prior to requesting the device token. If the received device certificate exists in the ban list **109**, the online service **101** notifies that host device **112**, and the connection between the host device **112** and remote accessory device **113** may be terminated immediately without continuing the remote authentication process (e.g., method **500**).

(37) From the perspective of the online service **400a**, in embodiments based on the first device token not existing within the token history (act **501**), act **507** comprises determining that the remote accessory device is counterfeit. As an example, the device token component **116** determines whether the first device token exists in the token history **108**. In embodiments, act **508** comprises adding the device certificate to a ban list. In embodiments, act **509** comprises sending an indication to the host device that the remote accessory device is banned based on the first device token not existing within the token history. As shown in act **510**, in some embodiments the host device terminates a connection with the remote accessory device based on receiving an indication from the online service.

(38) Embodiments of the disclosure may comprise or utilize a special-purpose or general-purpose computer system (e.g., online service **101**) that includes computer hardware, such as, for example, a processor system (e.g., processing system(s) **102**) and system memory (e.g., memory **103**), as discussed in greater detail below. Embodiments within the scope of the present disclosure also include physical and other computer-readable media for carrying or storing computer-executable instructions and/or data structures. Such computer-readable media can be any available media that

can be accessed by a general-purpose or special-purpose computer system. Computer-readable media that store computer-executable instructions and/or data structures are computer storage media (e.g., storage media **106**). Computer-readable media that carry computer-executable instructions and/or data structures are transmission media. Thus, by way of example, embodiments of the disclosure can comprise at least two distinctly different kinds of computer-readable media: computer storage media and transmission media.

(39) Computer storage media are physical storage media that store computer-executable instructions and/or data structures. Physical storage media include computer hardware, such as random access memory (RAM), read-only memory (ROM), electrically erasable programmable ROM (EEPROM), solid state drives (SSDs), flash memory, phase-change memory (PCM), optical disk storage, magnetic disk storage or other magnetic storage devices, or any other hardware storage device(s) which can be used to store program code in the form of computer-executable instructions or data structures, which can be accessed and executed by a general-purpose or special-purpose computer system to implement the disclosed functionality.

(40) Transmission media can include a network and/or data links which can be used to carry program code in the form of computer-executable instructions or data structures, and which can be accessed by a general-purpose or special-purpose computer system. A “network” is defined as one or more data links that enable the transport of electronic data between computer systems and/or modules and/or other electronic devices. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a computer system, the computer system may view the connection as transmission media. Combinations of the above should also be included within the scope of computer-readable media.

(41) Further, upon reaching various computer system components, program code in the form of computer-executable instructions or data structures can be transferred automatically from transmission media to computer storage media (or vice versa). For example, computer-executable instructions or data structures received over a network or data link can be buffered in RAM within a network interface module (e.g., network interface **104**), and then eventually transferred to computer system RAM and/or to less volatile computer storage media at a computer system. Thus, it should be understood that computer storage media can be included in computer system components that also (or even primarily) utilize transmission media.

(42) Computer-executable instructions comprise, for example, instructions and data which, when executed at one or more processors, cause a general-purpose computer system, special-purpose computer system, or special-purpose processing device to perform a certain function or group of functions. Computer-executable instructions may be, for example, binaries, intermediate format instructions such as assembly language, or even source code.

(43) It will be appreciated that the disclosed systems and methods may be practiced in network computing environments with many types of computer system configurations, including, personal computers, desktop computers, laptop computers, message processors, hand-held devices, multi-processor systems, microprocessor-based or programmable consumer electronics, network PCs, minicomputers, mainframe computers, mobile telephones, PDAs, tablets, pagers, routers, switches, and the like. Embodiments of the disclosure may also be practiced in distributed system environments where local and remote computer systems, which are linked (either by hardwired data links, wireless data links, or by a combination of hardwired and wireless data links) through a network, both perform tasks. As such, in a distributed system environment, a computer system may include a plurality of constituent computer systems. In a distributed system environment, program modules may be located in both local and remote memory storage devices.

(44) It will also be appreciated that the embodiments of the disclosure may be practiced in a cloud computing environment. Cloud computing environments may be distributed, although this is not required. When distributed, cloud computing environments may be distributed internationally within an organization and/or have components possessed across multiple organizations. In this description and the following claims, “cloud computing” is defined as a model for enabling on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services). A cloud computing model can be composed of various characteristics, such as on-demand self-service, broad network access, resource pooling, rapid elasticity, measured service, and so forth. A cloud computing model may also come in the form of various service models such as, for example, Software as a Service (SaaS), Platform as a Service (PaaS), and Infrastructure as a Service (IaaS). The cloud computing model may also be deployed using different deployment models such as private cloud, community cloud, public cloud, hybrid cloud, and so forth.

(45) Some embodiments, such as a cloud computing environment, may comprise a system that includes one or more hosts that are each capable of running one or more virtual machines. During operation, virtual machines emulate an operational computing system, supporting an operating system and perhaps one or more other applications as well. In some embodiments, each host includes a hypervisor that emulates virtual resources for the virtual machines using physical resources that are abstracted from view of the virtual machines. The hypervisor also provides proper isolation between the virtual machines. Thus, from the perspective of any given virtual machine, the hypervisor provides the illusion that the virtual machine is interfacing with a physical resource, even though the virtual machine only interfaces with the appearance (e.g., a virtual resource) of a physical resource. Examples of physical resources including processing capacity, memory, disk space, network bandwidth, media drives, and so forth.

(46) Although the subject matter has been described in language specific to structural features and/or methodological

acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the described features or acts described above, or the order of the acts described above. Rather, the described features and acts are disclosed as example forms of implementing the claims.

(47) The present disclosure may be embodied in other specific forms without departing from its essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

(48) When introducing elements in the appended claims, the articles “a,” “an,” “the,” and “said” are intended to mean there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Unless otherwise specified, the terms “set,” “superset,” and “subset” are intended to exclude an empty set, and thus “set” is defined as a non-empty set, “superset” is defined as a non-empty superset, and “subset” is defined as a non-empty subset. Unless otherwise specified, the term “subset” excludes the entirety of its superset (i.e., the superset contains at least one item not included in the subset). Unless otherwise specified, a “superset” can include at least one additional element, and a “subset” can exclude at least one element.

Claims

1. A method, implemented at a computer system that includes a processor, comprising: establishing an encrypted communication channel with a remote accessory device that is connected to a host device; receiving a first device token from the remote accessory device; determining that the first device token exists within a token history associated with a device certificate of the remote accessory device; determining that the remote accessory device is an authentic device, based on the first device token existing within the token history; sending an indication, to the host device, that the remote accessory device is authentic, based on the first device token existing within the token history; generating a second device token for the remote accessory device; storing the second device token in the token history associated with the device certificate of the remote accessory device; and sending the second device token to the remote accessory device.
2. The method of claim 1, further comprising determining the device certificate is valid.
3. The method of claim 1, the remote accessory device having been pre-authorized by the host device.
4. The method of claim 3, wherein the host device pre-authorizes the remote accessory device based only on communications between the remote accessory device and the host device.
5. The method of claim 1, wherein the host device is locally connected to the remote accessory device.
6. The method of claim 1, wherein the host device is remotely connected to the remote accessory device.
7. The method of claim 1, further comprising sending, to the remote accessory device, a request for the device certificate.
8. The method of claim 1, further comprising sending, to the remote accessory device, a request for an authentication token, wherein the first device token is received based on sending the request.
9. The method of claim 1, wherein the host device is a gaming console, and the remote accessory device is a gaming controller.
10. The method of claim 9, wherein the gaming console is a virtual gaming console.
11. A method, implemented at a computer system that includes a processor, comprising: establishing an encrypted communication channel with a remote accessory device that is connected to a host device; using the encrypted communication channel, obtaining a device certificate possessed by the remote accessory device; receiving a first device token from the remote accessory device; determining that the first device token does not exist within a token history associated with the device certificate; determining that the remote accessory device is counterfeit based on the first device token not existing within the token history; and adding the device certificate to a ban list.
12. The method of claim 11, further comprising, sending an indication, to the host device, that the remote accessory device is banned.
13. The method of claim 12, wherein, based on the indication, the host device terminates a connection with the remote accessory device.
14. The method of claim 11, wherein the remote accessory device was pre-authorized by the host device.
15. The method of claim 14, wherein the host device pre-authorizes the remote accessory device based only on communications between the remote accessory device and the host device.
16. The method of claim 11, further comprising requesting an authentication token from the remote accessory device, wherein the first device token is received based on requesting the authentication token.
17. The method of claim 11, wherein the host device is a gaming console, and the remote accessory device is a gaming controller.
18. A computer system comprising: a processing system; and a computer storage media that stores computer-executable instructions that are executable by the processing system to at least: obtain, over an encrypted network, a device certificate possessed by a remote gaming controller that is connected to a gaming console; receive a first device token from the remote gaming controller; determine that the first device token exists within a token history associated with the

device certificate; determine that the remote gaming controller is an authentic device based on the first device token existing within the token history; generate a second device token for the remote gaming controller; store the second device token in the token history associated with the device certificate; and send the second device token to the remote gaming controller.

19. The computer system of claim 18, the remote gaming controller having been pre-authorized by a host device.

20. The computer system of claim 18, wherein the computer-executable instructions that are executable by the processing system to, based on the first device token existing within the token history, send an indication, to the gaming console, that the remote gaming controller is authentic.
